Cross-Dataset Flow-Based Intrusion Detection: A Seven-Dataset Benchmark and Negative Results for Label-Free Transfer

Quang-Vinh Dang
British University Vietnam, Hung Yen, Vietnam
vinh.dq4@buv.edu.vn ORCID 0000-0002-3877-8024

Abstract

Machine-learning detectors of malicious network flows reach 0.985 ROC-AUC when trained and tested on one dataset and then lose most of that skill on another network. We build a benchmark of seven public flow datasets from five laboratories and three flow extractors (an eighth supports only a minimal feature tier), harmonised to 13 features of the same physical meaning, and evaluate eight baselines and nine variants of a method of our own in 42 ordered source–target transfers and seven leave-one-dataset-out (LODO) tasks over three seeds. Tree ensembles score 0.54 to 0.56 ROC-AUC on average across transfers, and 36 to 48% of the transfers are below chance; transfer within one laboratory is on average below chance too (0.44 for CatBoost) and strongly asymmetric. A domain-invariant method we designed, RIFT, is below its own gradient-boosting baseline (0.484 against 0.557 pairwise, 0.519 against 0.630 LODO). Giving RIFT the true target prior does not help (0.488), and also giving it a truly benign anchor recovers only part of the loss (0.543, still below the baseline); CORAL and DANN do not improve on boosting on average either. Transfer of a single attack family between datasets, benign against one family, is also near chance (median ROC-AUC 0.545 over 74 dataset–family cells, none reaching 0.9, against 0.999 in-domain), which excludes a difference of attack mix alone; no common feature keeps the sign of its association with the label (0 of 13 core features), and a classifier recognises the source dataset of a flow with 91.5% accuracy. The results are consistent with differences between datasets in the conditional distribution of the label, in benign background and in measurement that unlabelled adaptation cannot repair, but our controls cannot separate these, and our oracle search covers one informative pair. Source pre-training does not help the few-label case on average (0.835 against 0.928 with 0.1% of the target labels, 0.914 against 0.932 with matched trees). The package, configurations and all raw results are public.

1 Introduction

Flow-based network intrusion detection is usually evaluated by splitting one dataset into a training and a test part. Under that protocol gradient-boosted trees exceed 0.99 ROC-AUC on CIC-IDS2017 and USB-IDS-1. A growing literature shows that the same models degrade sharply on a second dataset [2, 5, 8, 9, 15], and even datasets from one laboratory transfer poorly. Two explanations are common. The first is covariate shift: capture tools, link speeds and traffic mixes change the scale and distribution of features, which normalisation or domain adaptation could repair [12, 16, 22]. The second is that the datasets define attacks differently, so that the same flow statistics carry different labels, which no unlabelled adaptation can repair.

This paper separates the two as far as a benchmark allows, and it reports a negative result in full. We make four contributions.

1. **A benchmark** of seven public datasets (an eighth, CTU-13, supports only a minimal feature tier; Table 1) with a common flow schema, a bottom-k hash sampler that gives a reproducible uniform sample of distinct rows, and a protocol with in-domain, pairwise (42 ordered pairs, labelled by whether the pair shares a laboratory or an extractor) and LODO settings, in inductive and transductive modes, over three seeds.

2. **A method designed to work, and its failure.** RIFT combines per-domain rank normalisation anchored on benign traffic, invariance-driven feature selection, and test-time label-shift correction. A full ablation and two oracle diagnostics (true target prior, truly benign anchor) show that it is below its own baseline and that supplying the information it estimates does not fix it.

3. **Controls** that answer the objection that a negative result may be an implementation failure, and that narrow down the explanation: positive controls, an oracle search over model family, normalisation and feature subsets (one informative pair), transfer of a single attack family between datasets, a nearest-neighbour test of whether the same flow statistics carry the same labels, a dataset-identity classifier, per-feature signs, and a few-label comparison. Table 7 lists what each control can and cannot show.

A companion study builds a few-label meta-learner on the diagnosis made here; this paper is self-contained.

2 Related work

Layeghy and Portmann [15] train on one NetFlow-v2 dataset and test on another and report attack-class F1 below 1% for several pairs of CIC-IDS2018, UNSW-NB15 and ToN-IoT, with strongly asymmetric results; a domain-invariance method of the same group [16] improves some pairs. Cantone et al. [5] report MCC between 0.26 and 0.44 when CIC-IDS2017 and CIC-IDS2018 are exchanged, even though both come from the same laboratory. D'hooge et al. [8, 9] analyse inter-dataset generalisation per attack class and the effect of data-damaged models on it. Catillo et al. [7] study transfer from CIC-IDS2017 to USB-IDS-1 [6], which share an extractor, and Apruzzese et al. [2] cross-evaluate per-attack detectors across several networks. Engelen et al. [11] document labelling errors in CIC-IDS2017, and Apruzzese et al. [3], Arp et al. [4] discuss pitfalls of machine-learning evaluation in security. Sarhan et al. [20] propose standard NetFlow feature sets to ease cross-dataset use. Our benchmark differs in covering seven datasets and three extractors with one schema, in labelling each pair by how much infrastructure the two datasets share, in running label-free transfer methods against default-setting baselines over three seeds, and in reporting the controls that decide whether the negative result is a finding. On the method side, we compare against domain-adversarial training [12] and correlation alignment [22], and RIFT uses the expectation-maximisation correction of the output prior of Saerens et al. [19].

3 Benchmark

Datasets. Table 1 lists the datasets: CIC-IDS2017 [21] and CIC-IDS2018, USB-IDS-1 [6], InSDN [10], UNSW-NB15 [17], ToN-IoT [1], IoT-23 [14] and CTU-13 [13]. They come from five laboratories (UNB, Sannio, UCD, UNSW Canberra and the Stratosphere laboratory of CTU) and three extractors (CICFlowMeter, Argus and Zeek). CIC-IDS2017 and CIC-IDS2018 share a laboratory and an extractor, UNSW-NB15 and ToN-IoT share a laboratory, and IoT-23 and CTU-13 share a laboratory.

Table 1: Datasets. The attack prior is that of the distinct rows after harmonisation. CTU-13 supports only the minimal feature tier and is excluded from the seven-dataset benchmark.

Dataset	Laboratory	Extractor	Attack prior	Feature tier
CIC-IDS2017	UNB CIC	CICFlowMeter	0.152	core
CIC-IDS2018	UNB CIC	CICFlowMeter	0.149	core
USB-IDS-1	Univ. of Sannio	CICFlowMeter	0.930	core
InSDN	UCD	CICFlowMeter	0.640	core
UNSW-NB15	UNSW Canberra	Argus, Bro	0.464	core
ToN-IoT	UNSW Canberra	Zeek	0.743	core
IoT-23	Stratosphere, CTU	Zeek	0.486	core
CTU-13	Stratosphere, CTU	Argus	0.570	minimal

The attack prior of the distinct rows ranges from 0.15 (CIC-IDS2017, CIC-IDS2018) to 0.93 (USB-IDS-1). Labels are binary, because attack taxonomies differ across datasets; the raw attack type is kept for per-family analysis. CTU-13 supports only a minimal feature tier and is therefore used only in the companion study, so the benchmark proper has seven datasets and 42 ordered pairs.

Common flow schema. Extractors export different features, so only features with the same physical meaning and units are used. The core tier has five base features (duration, forward and backward packets, forward and backward bytes) and eight derived rates and ratios (13 features), all computed after harmonisation and passed through $\log(1 + x)$. Ports and protocol are excluded because two of the datasets do not export them. Window-size and inter-arrival features exist only for some extractors and are excluded from the core tier. Alias tables resolve the different spellings of CICFlowMeter columns, repeated header rows of CIC-IDS2018 are removed, and negative sentinel values are clipped to zero.

Sampling. Each cleaned row receives a seeded 64-bit hash of all harmonised features the dataset provides (rounded to 10^{-6}) and its label, so distinctness is with respect to the full feature set of the dataset. Per label we keep the 100 000-row budget of rows with the smallest hash (a bottom-k sketch), so that duplicated flows occupy one slot and the sample is uniform over *distinct* rows. The attack prior is defined on distinct rows, and a class that is short shrinks the sample instead of being back-filled. This makes the benchmark a distinct-row benchmark, so absolute numbers are not comparable with studies that keep duplicates.

Protocol. Each dataset is split 70/30, stratified and seeded. Sources train on their training split and the target is evaluated on its test split. In the transductive mode a method also sees the unlabelled features of the target training split, disjoint from the evaluated rows; in the inductive mode it sees nothing from the target. The in-domain setting is the upper reference. The pairwise setting uses all 42 ordered pairs of the seven core datasets, each labelled *same family* (same laboratory, 4 pairs), *same extractor* (different laboratory, 12 pairs) or *cross-extractor* (26 pairs). In LODO a method trains on the other six datasets pooled. Target labels are never used for training, model selection or threshold tuning; threshold-dependent metrics use the decision rule of the method. The sample is drawn once; seeds 0, 1 and 2 change the 70/30 split and the model randomness, and we average seeds within a scenario before computing statistics over scenarios. Metrics are ROC-AUC (headline), PR-AUC, F1, balanced accuracy, MCC, recall and false-positive rate.

Methods. Baselines use a global standard scaler fitted on the sources: logistic regression, random forest (300 trees), histogram gradient boosting (HGB), CatBoost [18] with 500 iterations, XGBoost, and a two-layer MLP. We add CORAL [22] followed by HGB, and DANN [12]; both are transductive. RIFT is described next.

4 RIFT: a method designed to work

RIFT targets three sources of failure separately. *Covariate shift* is addressed by domain-wise normalisation: each domain is mapped to Gaussian ranks computed on its own features. On source domains the normaliser is fitted on benign rows only, because benign traffic is the stable background of a network. On the target, where labels are unknown, the anchor set is the $\gamma = 0.8$ fraction of the $(1 - \hat{\pi}_t)$ lowest-scoring rows under the current estimate $\hat{\pi}_t$ of the target attack prior, and the target is transported into the source representation by quantile matching against an equally truncated reference of source benign rows. *Spurious dataset-specific cues* are addressed by invariance-driven feature selection: a feature is kept only if the sign of its univariate association with the label is identical in every source and its minimum strength is at least 0.05; with a single source and a target pool (transductive mode), the 25% of features with the largest Kolmogorov–Smirnov distance between source and target are dropped, and without a target pool all features are kept; at least six features are always retained. *Label shift* is addressed by the expectation-maximisation correction of the output prior of Saerens et al. [19], applied after isotonic calibration. The classifier is HGB. Ablations remove one component at a time (domain normalisation, anchor, selection, prior correction), replace the rank transform by robust or z-score scaling, start the anchor from the pooled rather than the source normaliser, and add uncertainty-gated self-training.

5 Results

5.1 In-domain skill does not transfer

Table 2 gives ROC-AUC by setting. In-domain, the boosted ensembles reach 0.985 (random forest 0.981) averaged over the seven datasets (CatBoost between 0.950 on IoT-23 and 1.000 on ToN-IoT and USB-IDS-1), logistic regression 0.923. Across the 42 pairs the tree ensembles reach 0.54 to 0.56 (CatBoost 0.559, HGB 0.557, XGBoost 0.550, random forest 0.538) and DANN 0.565. Between 36% (HGB) and 48% (random forest) of the pairs are below 0.5 for the tree ensembles (21 to 29% are below 0.40), so on those pairs the learned rule ranks attacks as *less* suspicious than benign flows on the target. Only 3 of 42 pairs reach 0.9 for CatBoost, and all three involve USB-IDS-1 (USB-IDS-1→ToN-IoT 0.920, ToN-IoT→USB-IDS-1 0.948, InSDN→USB-IDS-1 0.959); the extreme pair is CIC-IDS2018→CIC-IDS2017 at 0.232. Pairs from the same laboratory, which are the easiest on paper, are on average the worst and are strongly asymmetric: CatBoost scores 0.44 on the four same-family pairs (CIC-IDS2017→CIC-IDS2018 0.75 but the reverse 0.23, ToN-IoT→UNSW-NB15 0.44, UNSW-NB15→ToN-IoT 0.34), 0.64 on same-extractor pairs and 0.54 on cross-extractor pairs; with four pairs and one favourable direction we do not test this ordering. LODO is better because six sources are pooled, but still 0.65 for CatBoost, and it varies from 0.968 on USB-IDS-1 to 0.333 on IoT-23 (Table 3).

No method is consistently best. A Friedman test over the 42 pairs finds a difference among the 17 transductive variants ($p = 0.006$), but that is driven by the weak methods (logistic regression and the RIFT variants); among the inductive methods ($p = 0.74$) and in LODO ($p = 0.065$ transductive, 0.47 inductive) it finds none. The pairwise differences among CatBoost, HGB, XGBoost and DANN

Table 2: ROC-AUC, mean over scenarios (seeds averaged within a scenario). Pairwise columns split the 42 ordered pairs by what the two datasets share. LODO: train on the other six datasets pooled. Transductive mode; the non-adaptive methods are unaffected by the mode up to run-to-run noise (at most 0.003).

Method	In-domain (7)	Same family (4 pairs)	Same extractor (12 pairs)	Cross-extractor (26 pairs)	LODO (7)
LR	0.923	0.341	0.472	0.503	0.480
RF	0.981	0.478	0.597	0.521	0.574
HGB	0.985	0.480	0.643	0.529	0.630
CatBoost	0.985	0.440	0.637	0.541	0.652
XGBoost	0.985	0.397	0.642	0.532	0.612
MLP	0.981	0.453	0.621	0.539	0.503
CORAL	0.985	0.494	0.494	0.542	0.631
DANN	0.980	0.516	0.678	0.521	0.613
RIFT	0.985	0.501	0.465	0.491	0.519
RIFT + self-training	0.985	0.519	0.448	0.486	0.500

Table 3: In-domain and leave-one-dataset-out ROC-AUC per target (inductive).

Target	In-domain	LODO CatBoost	LODO HGB
CIC-IDS2017	0.999	0.803	0.775
CIC-IDS2018	0.970	0.656	0.684
InSDN	0.998	0.672	0.657
IoT-23	0.950	0.333	0.308
ToN-IoT	1.000	0.667	0.560
UNSW-NB15	0.978	0.468	0.467
USB-IDS-1	1.000	0.968	0.957

are of the order of 0.01 (in LODO up to 0.04, with CatBoost best on mean rank), far below the spread between pairs; with 7 LODO targets and pairs that share sources and targets, the Friedman tests have little power, so the data neither show nor exclude a best method. On average CORAL and DANN do not repair transfer: pairwise they are at 0.524 and 0.565, in LODO at 0.631 and 0.613, against 0.630 for HGB, although both differ strongly by target (on LODO IoT-23 CORAL reaches 0.72 and DANN 0.76 against 0.31 for HGB).

5.2 RIFT fails

Table 4 shows RIFT and its ablations. RIFT is below its own HGB baseline in both settings (pairwise 0.484 against 0.557, LODO 0.519 against 0.630). Removing domain normalisation recovers the baseline (0.558 pairwise, 0.620 LODO), and removing the benign anchor recovers part of it (0.538, 0.543), so the target-side normalisation and the benign anchor are what hurt. Removing the label-shift correction changes nothing (0.484 against 0.484 pairwise), as it must for ROC-AUC, because a prior-ratio adjustment is a monotone transform of the posterior. In LODO the invariant selection has no effect either, for a mechanical reason: no feature has the same sign in all six sources (consistent with Sect. 6), so the selection falls back to keeping all 13 features in every LODO scenario; in the pairwise setting the Kolmogorov–Smirnov rule keeps 10 of 13. Replacing the rank transform by robust or z-score scaling does not help (0.479 and 0.485 pairwise). Self-training does not help (0.478). In the inductive pairwise setting there is no target pool and a single source, so RIFT keeps all features and reduces to HGB on a monotone rank transform; all variants score

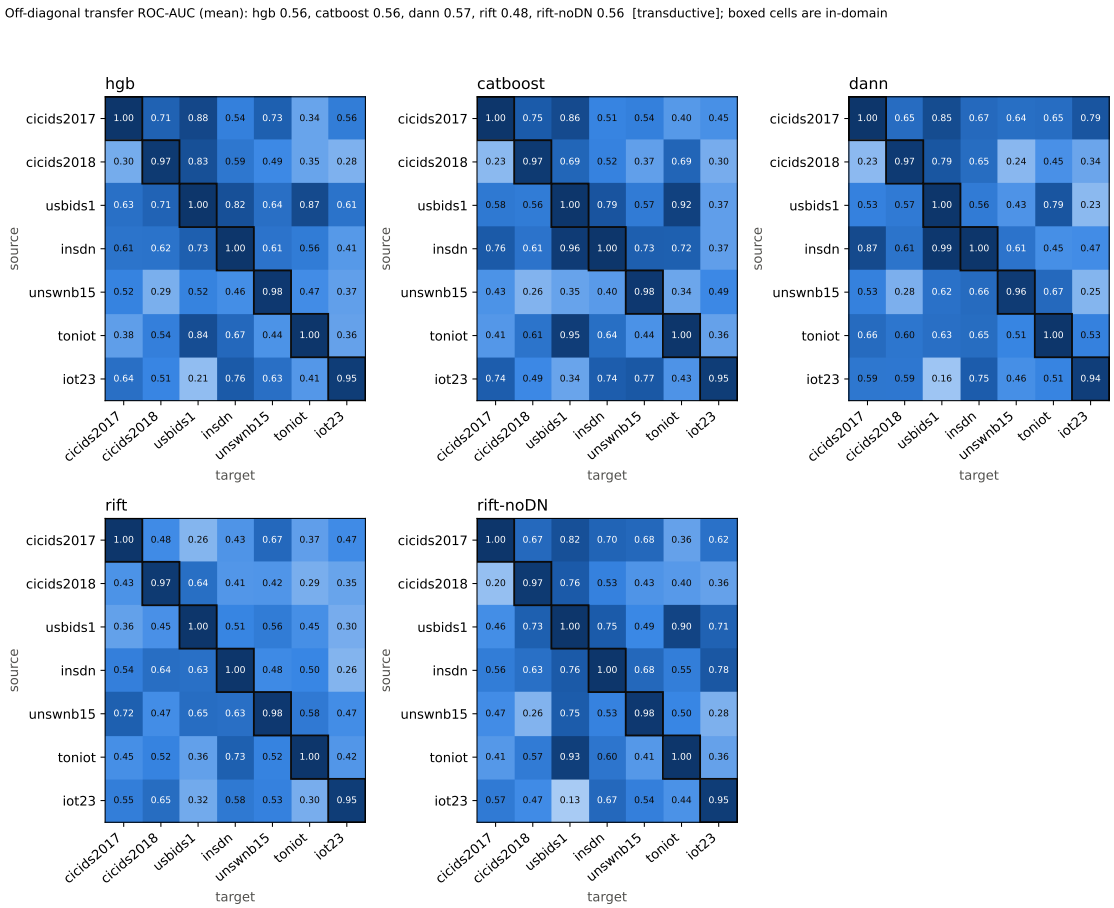


Figure 1: Source→target ROC-AUC of the benchmark methods (transductive mode); the boxed diagonal is the in-domain value.

the same (0.551 against 0.554 for HGB). In inductive LODO the per-domain source normalisation and the multi-source selection remain active and the variants differ (RIFT 0.586, without domain normalisation 0.620, robust scaling 0.502); RIFT is below HGB there too (0.630), and removing domain normalisation raises the score without any target data.

The target-prior estimate is poor, although removing the correction does not matter for ROC-AUC and the prior error is similar for the variants that recover the baseline (0.39 and 0.38 for the variants without domain normalisation and without anchor against 0.41 for RIFT), so the error alone does not explain the loss. Across the 42 pairs RIFT estimates the target attack prior with a mean absolute error of 0.41 (bias -0.35, Pearson correlation with the true prior 0.05); in LODO it estimates an average prior of 0.003 where the true average is 0.51. The anchor is meant to be benign traffic, but its purity (the fraction of truly benign rows, logged as a diagnostic and never used by the method) is 0.50 in pairwise and 0.47 in LODO transfer, so the anchor is as contaminated with attacks as the pool. Estimating the prior with a normaliser fitted on the pool ("pooled init.") does not help (0.487). We hypothesise, without having tested it, that a source classifier that is inverted on the target gives the EM nothing to correct.

Oracle diagnostics. Two diagnostics read the labels of the target pool and are therefore not label-free methods. Supplying the true target prior to the anchor and to the correction gives 0.488

Table 4: RIFT and its ablations, transductive mode. Pairwise: 42 pairs; LODO: 7 targets. Prior error (pairwise only): mean absolute error of the estimated target attack prior.

Method	Pairwise AUC	Pairwise bal. acc.	LODO AUC	Prior error
HGB	0.557	0.533	0.630	–
RIFT	0.484	0.510	0.519	0.413
RIFT − domain norm.	0.558	0.536	0.620	0.385
RIFT − benign anchor	0.538	0.521	0.543	0.381
RIFT − invariant selection	0.483	0.495	0.519	0.429
RIFT − prior correction	0.484	0.509	0.519	–
RIFT, pooled init.	0.487	0.500	0.498	0.421
RIFT, robust scaling	0.479	0.504	0.487	0.425
RIFT, z-score scaling	0.485	0.498	0.510	0.414
RIFT + self-training	0.478	0.507	0.500	0.409
RIFT, true prior (diagnostic)	0.488	0.503	0.562	0.000
RIFT, true prior and benign anchor (diagnostic)	0.543	0.507	0.596	0.000

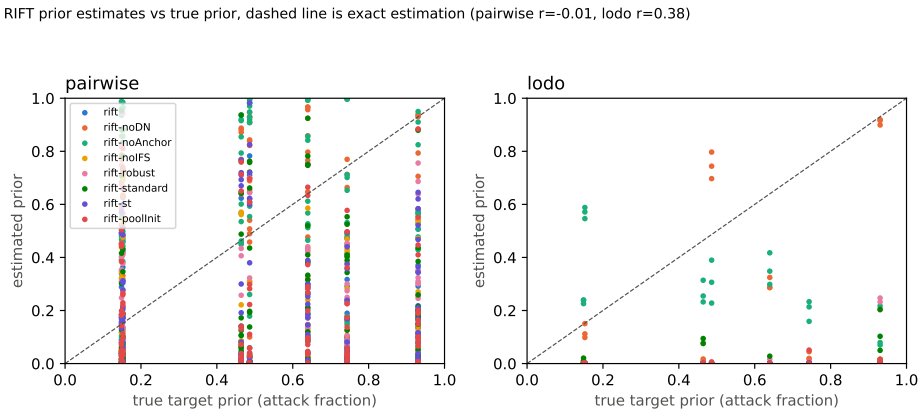


Figure 2: Estimated against true target attack prior for the RIFT variants.

pairwise and 0.562 LODO, and the anchor stays as contaminated (purity 0.48 and 0.49), so a perfect prior estimate does not repair RIFT. Additionally replacing the anchor by truly benign pool rows gives 0.543 and 0.596, which recovers most of RIFT's loss but stays below HGB (0.557 and 0.630), and balanced accuracy stays at 0.51. The benign anchor therefore matters, but with perfect information the target-side normalisation still does not improve on no adaptation, which is consistent with the failure lying in the rule that the source classifier carries and not in the estimation of the shift.

5.3 Source data do not help in the few-label setting

As a reference outside the label-free benchmark, we continue a source-pretrained CatBoost on a labelled budget of 0.1%, 1% and 5% of the target training split (on average 62, 617 and 3 083 rows), and compare it with CatBoost trained on those rows alone. At 0.1%, the source-pretrained model reaches 0.835 and the target-only model 0.928 (Table 5). Giving the continued model 300 or 500 extra trees narrows the gap to 0.911 and 0.914 against 0.932, so the effect is not only a capacity artefact, but it does not reverse: at 1% and 5% the two are within 0.001 to 0.002 once the trees are matched. Source pre-training thus gives no average benefit over target labels for this model family, and a harm at the smallest budget (0.09 AUC with 100 trees, 0.02 with 500); continuing a

Table 5: Few-label comparison over the 42 pairs, ROC-AUC. The continued model starts from a CatBoost trained on the source and adds the stated number of trees on the labelled target rows; the other model uses only those rows.

Setting	Extra trees	0.1% labels	1% labels	5% labels
Source-pretrained, continued	100	0.835	0.950	0.966
	300	0.911	0.966	0.978
	500	0.914	0.966	0.978
Target labels only	100	0.928	0.964	0.977
	300	0.932	0.967	0.980
	500	0.932	0.967	0.979

boosted model by adding trees is a weak way to use source data, and stronger few-label baselines (pooled training with target weighting, source scores as a feature, an in-context model given the source rows) are not part of this study.

5.4 Shift and family explanations

An unsupervised shift score (an isolation forest fitted on the source and applied to the target pool) does not predict transfer skill: the Spearman correlation with zero-shot ROC-AUC over 2 142 (scenario, method, seed) points is -0.055 (nominal $p = 0.01$, but the points are not independent; per method $|\rho| \leq 0.27$ with mixed signs). Sharing attack types does not obviously predict it either: for every family we compared the informedness of a method on a target family when the source contains that family and when it does not. Pooled over families (transductive mode), the paired difference for CatBoost is $+0.012$ (95% half-width 0.088, $p = 0.82$) and for HGB -0.046 ($p = 0.14$); none of the 248 (method, mode, family) tests has $p < 0.05$, but each test uses only 3 to 6 paired cells and cannot reach 0.05 with fewer than 6, so the benchmark has too little power to show whether missing attack types matter.

6 Controls

A negative result is a finding only if competing explanations are ruled out.

The pipeline works. The same pipeline reaches 0.985 in-domain, and a target-only CatBoost with 0.1% of target labels reaches 0.93, so the features carry the signal and the code can learn it.

Tuning as an explanation (one informative pair). For CIC-IDS2017→CIC-IDS2018 we searched all five model families (HGB, CatBoost, XGBoost, random forest, logistic regression) with two settings of the main hyperparameters, three normalisations (none, global, per-domain rank) and the full feature set or the set without one feature group (264 configurations, seed 0, 30 000 source rows), and selected the best on the target test split, an optimistic oracle. It reaches 0.799 against 0.683 for the benchmark HGB, still far from the in-domain 0.97 to 1.00. For CIC-IDS2017→USB-IDS-1 only 33 HGB configurations were run and the oracle (0.862) is below the full-data benchmark HGB (0.972), so that pair says nothing about tuning. The grid is reduced and the source is subsampled, so the result is not an upper bound over all configurations.

Table 6: Transfer of a single attack family between datasets: HGB trained on benign plus the family in the source dataset, ROC-AUC on benign plus the family in the target dataset (seed 0, up to 20 000 rows per class).

Family	Cells	Median cross-dataset AUC	Min	Max	Median in-domain AUC
DoS	30	0.569	0.313	0.857	0.999
DDoS	12	0.526	0.245	0.750	1.000
Scan	12	0.598	0.120	0.900	1.000
Brute force	12	0.514	0.148	0.829	0.999
Exploit / malware	6	0.543	0.464	0.814	0.993
Botnet	2	0.366	0.042	0.689	0.918
All	74	0.545	0.042	0.900	1.000

The same statistics do not carry the same labels. We standardise source and target with source statistics, find for each target row its 10 nearest source rows, and restrict to the *overlap region* (rows whose mean neighbour distance is at most the 90th percentile of source-to-source distances). There the majority label of the neighbours should agree with the true target label if $P(y \mid x)$ were shared. Over the 19 pairs with at least 100 overlapping rows (seed 0; 23 of the 42 pairs have fewer) the median Cohen's κ is 0.03, 84% of the pairs have $\kappa \leq 0.1$ and the maximum is 0.25 (Fig. 3); three of the 19 values are exactly 0, which a single-class overlap region also produces, and only five are negative, so the pattern is closer to independence than to inversion. The in-domain reference, source-test rows against the same source-train index, has κ of 0.91 on average and is not matched in region or difficulty. The pair CIC-IDS2017→CIC-IDS2018, from one laboratory, has $\kappa = -0.07$ with 40% of the target rows in the overlap region.

The same attack family does not transfer either. For every attack family present in at least two core datasets with at least 200 rows each (DoS, DDoS, scan, brute force, exploit/malware and botnet; families assigned by name rules in `crossids.analysis`) we trained HGB on benign flows plus that family in one dataset and scored benign flows plus that family in another: 74 ordered dataset–family cells with up to 20 000 rows per class (Table 6). The median cross-dataset ROC-AUC is 0.545, 36% of the cells are below 0.5, 7% reach 0.8 and none reaches 0.9, whereas the in-domain version of the same task has median 1.000 (minimum 0.82). The attack mix is thus held fixed up to the coarse family and transfer still fails, which excludes a difference of attack mix alone. It does not separate a change of the conditional label distribution from differences in benign traffic and in how flows are measured, since benign flows also differ between datasets (below); the family assignment is coarse (DoS covers different tools), and the botnet family has two cells.

No flow feature keeps its meaning. For each feature we computed the univariate ROC-AUC with the label in every dataset. The sign of the association (above or below 0.5) is mixed for all 14 features of the eight datasets (13 core features and the minimal-tier `bytes_per_pkt`) and also for all 13 core features of the seven core datasets, with per-dataset AUC ranges such as 0.025 to 0.760 for total bytes and 0.057 to 0.808 for forward packets (Fig. 4).

Datasets are easy to tell apart. A CatBoost classifier recognises the dataset of a flow among the seven core datasets with 91.5% accuracy (chance 14.3%), and 91.8% on benign flows alone. Benign traffic therefore also differs between datasets, which is consistent with differences in capture setup and background traffic.

Table 7: What each control can and cannot show.

Control	Rules out	Cannot rule out
In-domain skill (0.985)	A broken pipeline or uninformative features	Anything about transfer; in-domain tasks may be easy for testbed-generated attacks
Oracle search (one informative pair)	Default hyperparameters of HGB on that pair	Tuning in general (one pair, reduced grid, subsampled source)
Single-family transfer	A difference of attack mix alone	Differences in benign traffic, in measurement, and in the conditional label distribution
Nearest-neighbour κ	Nothing alone: the overlap is small and $\kappa = 0$ also arises from a single-class region	Shift in $P(y \mid x)$; the test has little power
Per-feature signs	A stable direction of any single feature	Attack-mix shift (no within-dataset null)
Dataset identity (91.5%)	Datasets being interchangeable	Which kind of shift it is; it indicates large covariate or support shift
RIFT oracle diagnostics	Prior estimation as the cause of RIFT's failure	Other adaptation designs
Few-label comparison	Source pre-training by continuing boosted trees as useful on average	Better ways to use source data

Published results agree. The reported collapses in the literature occur in the same regime: attack-class F1 below 1% for several NetFlow-v2 pairs [15], MCC between 0.26 and 0.44 for CIC-IDS2017 and CIC-IDS2018 [5]. Metrics and feature sets differ across studies, so we compare the pattern and not the numbers.

7 Discussion

The benchmark shows that cross-dataset transfer fails on average with every method we ran, though not on every pair: three to five pairs exceed 0.8 per tree method, mostly with USB-IDS-1 as source or target. It is not repaired by the usual remedies: default-setting tuning on the pair we tested, normalisation, feature selection, alignment, adversarial invariance, prior correction and self-training, and RIFT is not rescued by the true prior or a truly benign anchor. Below-chance ROC-AUC on 36 to 48% of the pairs (21 to 29% below 0.40) for the tree ensembles is what an inverted rule would produce, but also what a different attack mix would produce; the single-family transfer, with the attack mix held fixed up to the family, is also near chance (median 0.545), which excludes attack mix alone. What remains is a difference in benign traffic, in how flows are measured, or in the conditional label distribution, and the controls cannot tell which.

What the evidence does not show. First, the per-feature sign flips arise when the attack mix differs, even without a labelling disagreement, and we have no within-dataset null for them. The nearest-neighbour test conditions on flow statistics but is restricted to the overlap region, which is small for most pairs (a median overlap fraction below 0.01), and a κ near zero there means only that the neighbours' labels are uninformative; three of the 19 values are exactly zero. Second, "labelling function" is not the right name for what the benchmark can establish: with only 13 flow features and no ports, flags or payload, $P(y \mid x)$ also changes when relevant variables are omitted and their distribution differs between datasets, even if every dataset applied the same rule. The datasets label

attacks by different procedures (attacker address and schedule, scenario, tool logs), which we did not tabulate. Third, the oracle control covers one informative pair, and the single-family control uses a coarse family assignment and cannot separate benign and measurement differences from the conditional label distribution. Fourth, the seven datasets are public laboratory captures with their own labelling errors [11], the flow features come from different extractors whose byte and duration semantics are not audited here, and the distinct-row sampling changes attack priors relative to natural traffic; the features and sampling are not varied. Fifth, RIFT is one design; the negative result concerns this design and these ablations, and the baselines run with default settings.

Implications. Evaluation should report cross-dataset results with a pair taxonomy and per-pair distributions, because same-laboratory pairs can be the worst, one favourable pair hides the rest, and a mean of values near 0.2 and 0.97 says little; a method should be compared with its own baseline in the same setting, as here, because it can otherwise look useful for the wrong reason. Methods that assume a fixed $P(y \mid x)$ should not be expected to repair these transfers, and the few-label results suggest that a few labelled flows of the target network are worth more than the source data on average, for the model family we tested.

8 Conclusion

On seven public datasets, flow classifiers that reach 0.985 ROC-AUC in-domain reach 0.54 to 0.57 across datasets, with about 40% of the transfers below chance for the tree ensembles, and transfer within a laboratory is on average worse than across laboratories. A method designed for the covariate and label shifts of this setting (RIFT) is below its own baseline, and neither the true target prior nor a truly benign anchor makes it better than no adaptation. Transfer of a single attack family between datasets is also near chance, so a difference of attack mix alone does not explain the failure; the controls are consistent with differences in benign traffic, measurement and the conditional label distribution, which they cannot separate. The practical consequence is that a handful of labelled flows from the target network is the reliable ingredient.

Reproducibility. The package, configurations, sampled-row hashes and every raw per-run result are in `https://github.com/vinhqdang/cross_ids`. Tables and figures are generated from the result files by the repository scripts.

References

[1] Abdullah Alsaedi, Nour Moustafa, Zahir Tari, Abdun Mahmood, and Adnan Anwar. TON_IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems. *IEEE Access*, 8:165130–165150, 2020.

[2] Giovanni Apruzzese, Luca Pajola, and Mauro Conti. The cross-evaluation of machine learning-based network intrusion detection systems. *IEEE Transactions on Network and Service Management*, 19(4):5152–5169, 2022. doi: 10.1109/TNSM.2022.3157344. arXiv:2203.04686.

[3] Giovanni Apruzzese, Pavel Laskov, and Johannes Schneider. SoK: Pragmatic assessment of machine learning for network intrusion detection. In *2023 IEEE 8th European Symposium on Security and Privacy (EuroS&P)*, pages 592–614, 2023. doi: 10.1109/EuroSP57164.2023.00042.

[4] Daniel Arp, Erwin Quiring, Feargus Pendlebury, Alexander Warnecke, Fabio Pierazzi, Christian Wressnegger, Lorenzo Cavallaro, and Konrad Rieck. Dos and don'ts of machine learning in computer security. In *31st USENIX Security Symposium (USENIX Security 22)*, pages 3971–3988, 2022. arXiv:2010.09470.

[5] Marco Cantone, Claudio Marrocco, and Alessandro Bria. Machine learning in network intrusion detection: A cross-dataset generalization study. *IEEE Access*, 12:144489–144508, 2024. doi: 10.1109/ACCESS.2024.3472907. arXiv:2402.10974, titled "On the Cross-Dataset Generalization of Machine Learning for Network Intrusion Detection".

[6] Marta Catillo, Andrea Del Vecchio, Luciano Ocone, Antonio Pecchia, and Umberto Villano. USB-IDS-1: a public multilayer dataset of labeled network flows for IDS evaluation. In *2021 51st Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*, pages 1–6, 2021. doi: 10.1109/DSN-W52860.2021.00012.

[7] Marta Catillo, Andrea Del Vecchio, Antonio Pecchia, and Umberto Villano. Transferability of machine learning models learned from public intrusion detection datasets: the CICIDS2017 case study. *Software Quality Journal*, 30(4):955–981, 2022. doi: 10.1007/s11219-022-09587-0.

[8] Laurens D'hooge, Tim Wauters, Bruno Volckaert, and Filip De Turck. Inter-dataset generalization strength of supervised machine learning methods for intrusion detection. *Journal of Information Security and Applications*, 54:102564, 2020. doi: 10.1016/j.jisa.2020.102564.

[9] Laurens D'hooge, Miel Verkerken, Tim Wauters, Filip De Turck, and Bruno Volckaert. Characterizing the impact of data-damaged models on generalization strength in intrusion detection. *Journal of Cybersecurity and Privacy*, 3(2):118–144, 2023. doi: 10.3390/jcp3020008.

[10] Mahmoud Said Elsayed, Nhien-An Le-Khac, and Anca Delia Jurcut. InSDN: A novel SDN intrusion dataset. *IEEE Access*, 8:165263–165284, 2020.

[11] Gints Engelen, Vera Rimmer, and Wouter Joosen. Troubleshooting an intrusion detection dataset: the CICIDS2017 case study. In *2021 IEEE Security and Privacy Workshops (SPW)*, pages 7–12, 2021. doi: 10.1109/SPW53761.2021.00009.

[12] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016.

[13] Sebastián García, Martin Grill, Jan Stiborek, and Alejandro Zunino. An empirical comparison of botnet detection methods. *Computers & Security*, 45:100–123, 2014.

[14] Sebastian Garcia, Agustin Parmisano, and Maria Jose Erquiaga. IoT-23: A labeled dataset with malicious and benign IoT network traffic, 2020. Stratosphere Laboratory, Czech Technical University in Prague. Zenodo, doi:10.5281/zenodo.4743746.

[15] Siamak Layeghy and Marius Portmann. Explainable cross-domain evaluation of ML-based network intrusion detection systems. *Computers and Electrical Engineering*, 108:108692, 2023. doi: 10.1016/j.compeleceng.2023.108692. arXiv:2205.04112, titled "On Generalisability of Machine Learning-based Network Intrusion Detection Systems".

[16] Siamak Layeghy, Mahsa Baktashmotlagh, and Marius Portmann. DI-NIDS: Domain invariant network intrusion detection system. *Knowledge-Based Systems*, 273:110626, 2023. doi: 10.1016/j.knosys.2023.110626. arXiv:2210.08252.

[17] Nour Moustafa and Jill Slay. UNSW-NB15: A comprehensive data set for network intrusion detection systems. In *Military Communications and Information Systems Conference (MilCIS)*, 2015.

[18] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems 31*, 2018.

[19] Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. *Neural Computation*, 14(1):21–41, 2002. doi: 10.1162/089976602753284446.

[20] Mohanad Sarhan, Siamak Layeghy, and Marius Portmann. Towards a standard feature set for network intrusion detection system datasets. *Mobile Networks and Applications*, 27(1):357–370, 2022. doi: 10.1007/s11036-021-01843-0.

[21] Iman Sharafaldin, Arash Habibi Lashkari, and Ali A. Ghorbani. Toward generating a new intrusion detection dataset and intrusion traffic characterization. In *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP)*, pages 108–116, 2018.

[22] Baochen Sun, Jiashi Feng, and Kate Saenko. Return of frustratingly easy domain adaptation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2016.

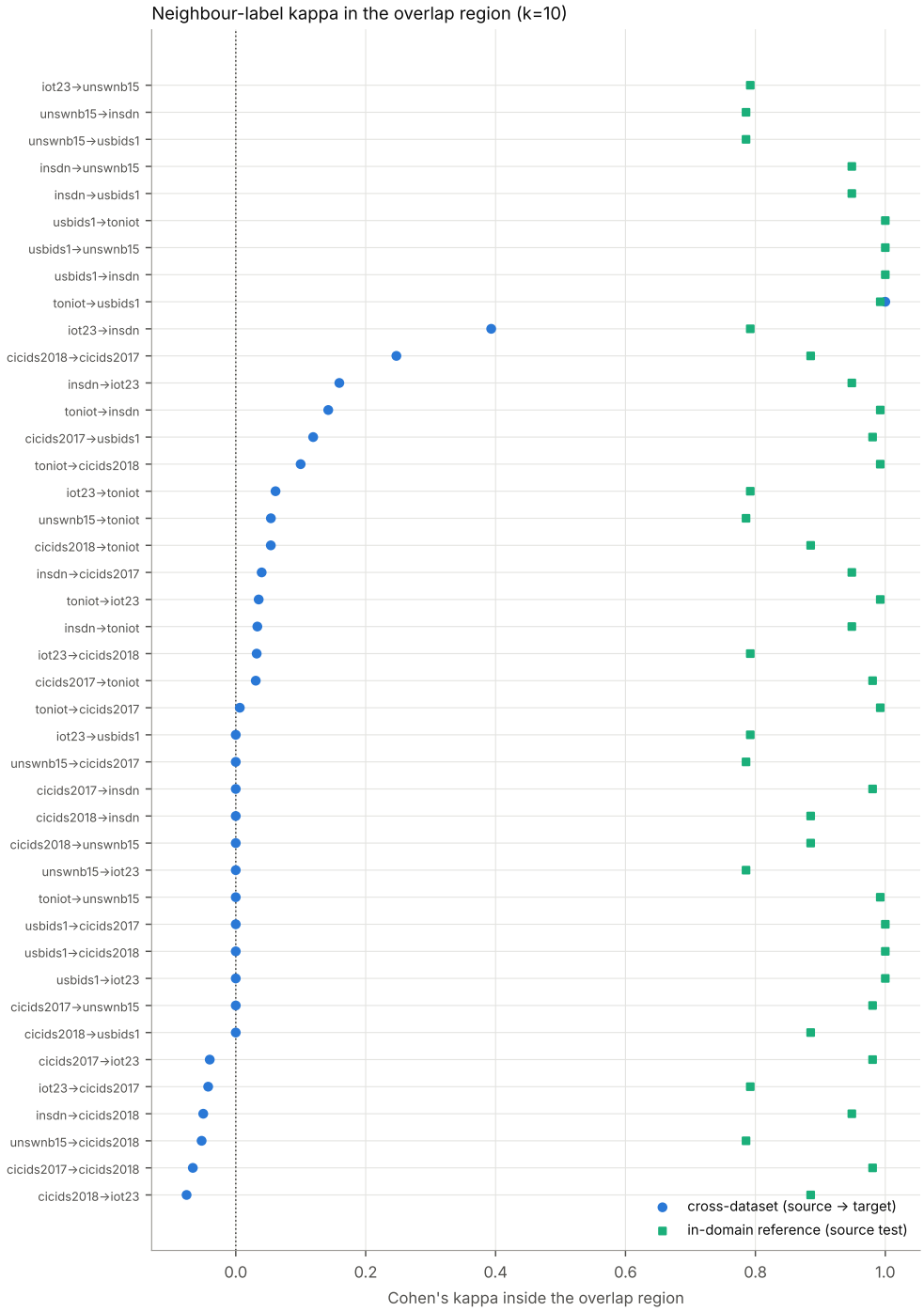


Figure 3: Cohen's κ between the majority label of the nearest source rows and the true target label in the overlap region, per pair, against the in-domain reference.

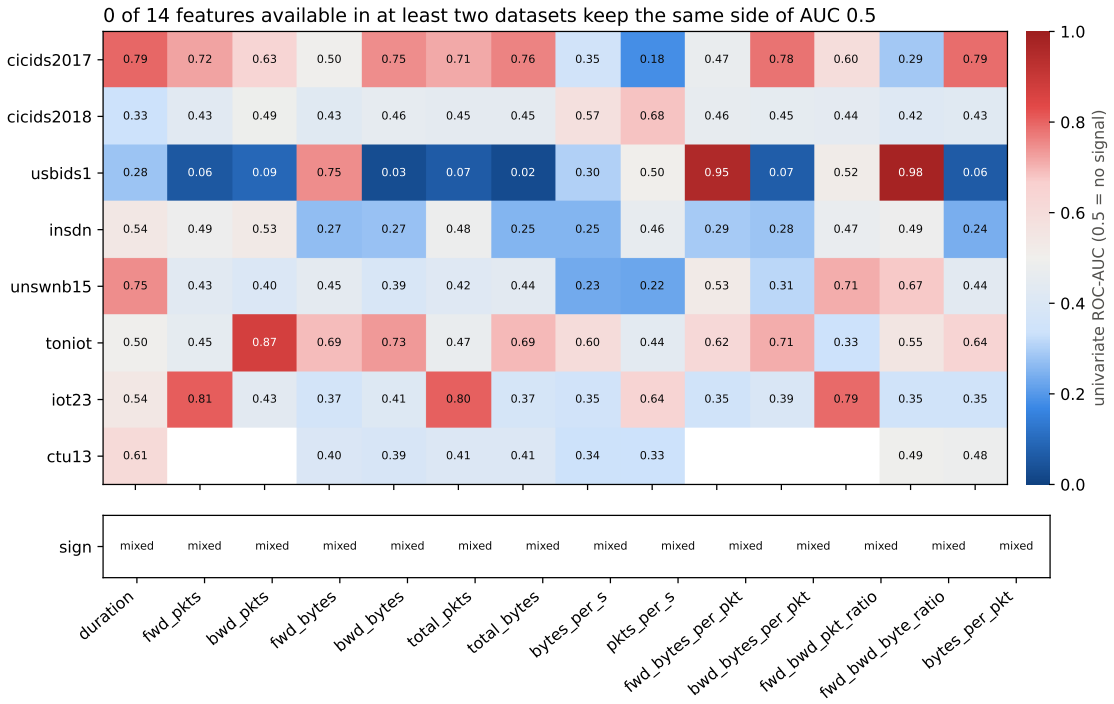


Figure 4: Univariate ROC-AUC of each core feature with the attack label, per dataset (centred at 0.5); the figure includes CTU-13 and the minimal-tier feature `bytes_per_pkt`.